CECS551 Final Project - Phase 3 Report
Preliminary Data Analysis and Feasibility

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Dataset plan

Source and Access Method

The dataset is from the Enveda CASMI 2026 - Molecule ID From Mass Spectra | Kaggle competition. The data will be accessed by joining the Kaggle competition and using both Kaggle notebooks and the Kaggle API. Kaggle notebooks will be used for model training and inference, taking advantage of the available free weekly GPU and TPU resources. The Kaggle API will be configured using an API key generated through the Kaggle account to access competition files from external environments when needed.

Expected Size and Labels

The dataset contains mass spectrometry data paired with molecular structures represented as SMILES strings. The training set includes approximately 2.5 million MS/MS spectra from 275,000 unique molecules, with one row per spectrum. The test set contains approximately 1,500 spectra from 400 molecules. The model will use spectra as inputs and SMILES strings as target labels to predict molecular structures.

Realistic Scope

The dataset has been used in other Kaggle notebooks for predicting molecular structures given mass spectra, so it should be of sufficient quality to train a model.

Although the dataset is over 3GB in size and contains only a few examples per molecule, it is realistic to train from because: similar motifs exist between different molecules, so though there are few examples of each molecule, the fragmentation patterns of the motifs appear repeated between different molecules for the model to learn, also Database Retrieval & analog mass spectra shifting can be used to generate additional examples for the model to train from.
This has been shown by other notebooks in the competition and by methods such as MSNovelist - https://www.nature.com/articles/s41592-022-01486-3

Preliminary EDA

This section provides information on the dataset that can help determine how model parameters should be set to produce good predictions from the input data.

Table 1: Guide to the plots and tables in this section

Figure	What it shows	Why it matters for modelling
1	Precursor m/z vs. peaks per spectrum (heatmap with marginal histograms)	shows which mass ranges dominate and how complex spectra are
2	Precursor m/z vs. total spectrum abundance (bubble size and color = fragment count)	Exposes low-signal spectra and outliers
3	Adduct distribution (counts and percentages)	Class balance, needed to stratify folds evenly across positive and negative ion modes
4	Number of spectra per unique molecule	Measures data replication and per-structure coverage
5	Interactive 3D structure beside its MS2 spectrum	Sample inspection, checks peaks against structure
6	Kendrick mass defect (CH_2 base) of fragment ions	Shows the chemical structure of the fragment population
7	Fractional mass defect of the precursor vs. m/z	Flags unusual composition (e.g., heavy atoms)
8	SMILES string length vs. precursor mass	Informs decoder sequence length
9	Ring and branch token counts vs. precursor mass	Shows structural complexity at high mass
10	Intensity retained vs. encoder input length	Sets the encoder input budger
11	SMILES token-length distribution and truncation rate	Sets the decoder length limit

1. Dataset Composition and Class Balance

1.1 Precursor Mass and Spectrum Complexity

Precursor m/z values are mostly around 300–350 and have a long right tail past 1000 (Figure 1). Most spectra contain fewer than 20 peaks, while only a few have up to 65. These larger spectra are more sensitive to the encoder's input length. Most spectra also have low total abundance, with a few higher-abundance spectra between 350 and 900 m/z (Figure 2).

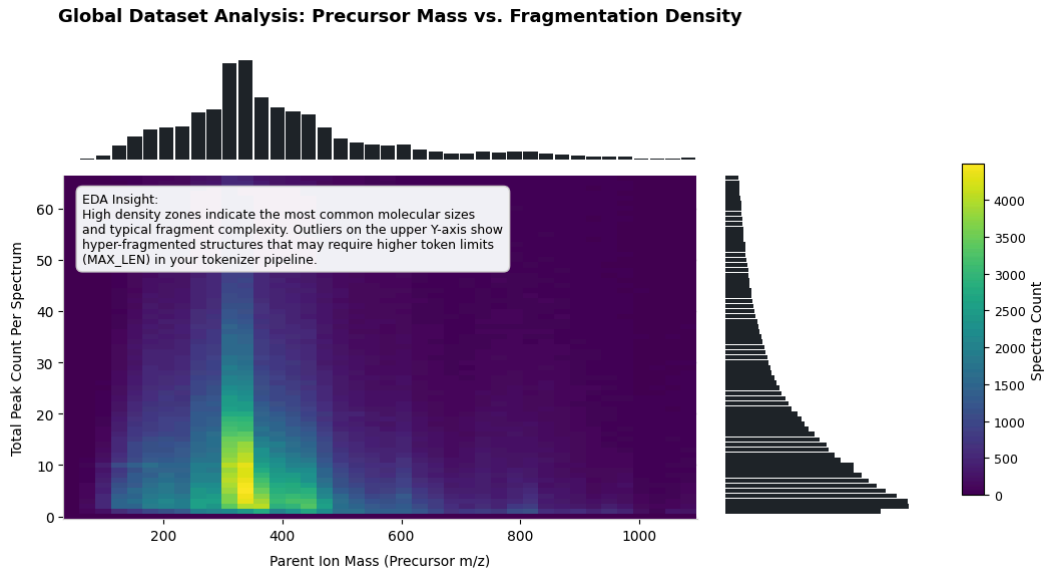


Figure 1. Precursor m/z vs. total peak count per spectrum. A depiction of how many fragments are expected to result from original molecules of a given mass.

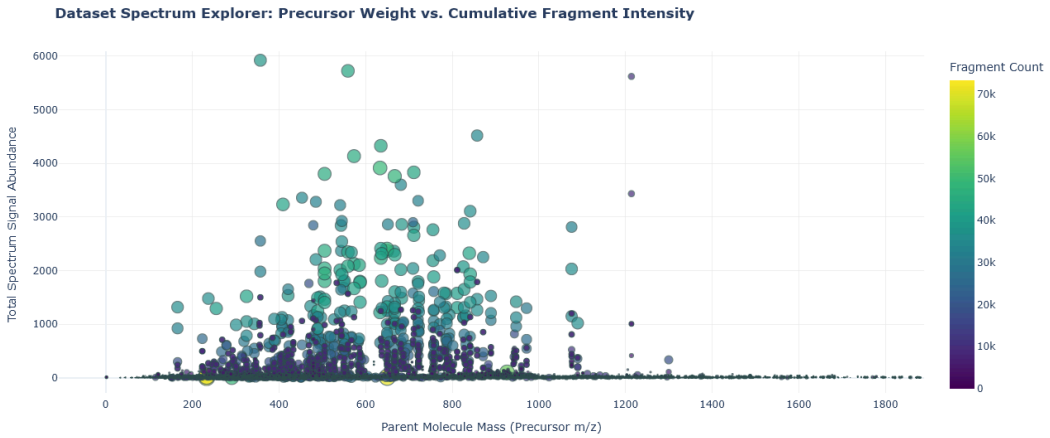


Figure 2. Precursor m/z vs. cumulative fragment intensity per spectrum. Another view of the distribution of how many fragments result from different initial molecule masses.

1.2 Adduct Balance

An adduct forms when a molecule attaches to an ion, such as a proton or sodium ion, during mass spectrometry. It causes a shift in the mass of the molecule and polarity of charge. This must be identified and accounted for when identifying the original molecule.

The adduct distribution is highly imbalanced (Figure 3). [M+H]+ accounts for 53.1% of spectra followed by [M-H]- at 20.3% and [M+Na]+ at 11.7%. The remaining adducts together make up about 15%, and about seven of them are at or below 0.2%. Roughly three-quarters of spectra are positive mode. This supports stratifying cross-validation folds by adduct and reporting per-adduct metrics, since rare adducts cannot be evaluated reliably.

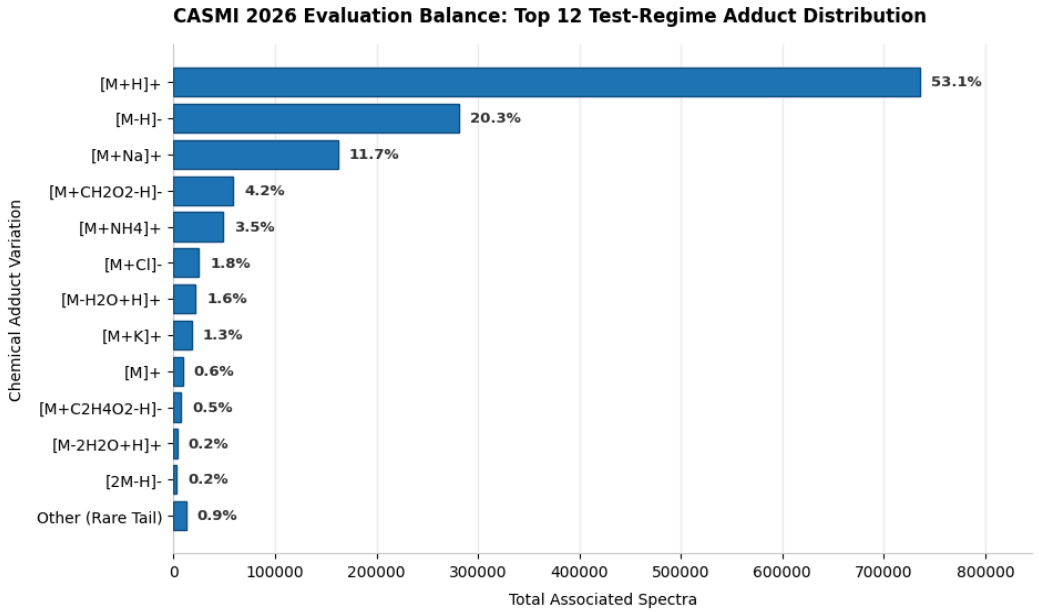


Figure 3. Adduct distribution in the evaluation set.

1.3 Redundancy Across Molecules

Spectra are unevenly spread across structures (Figure 4). About 15,000 molecules have a single spectrum and the median is 7 spectra per molecule. Molecules with only one or a few examples may make it hard for a model to separate noise from the underlying fragmentation chemistry, while heavily replicated molecules can dominate the training loss.

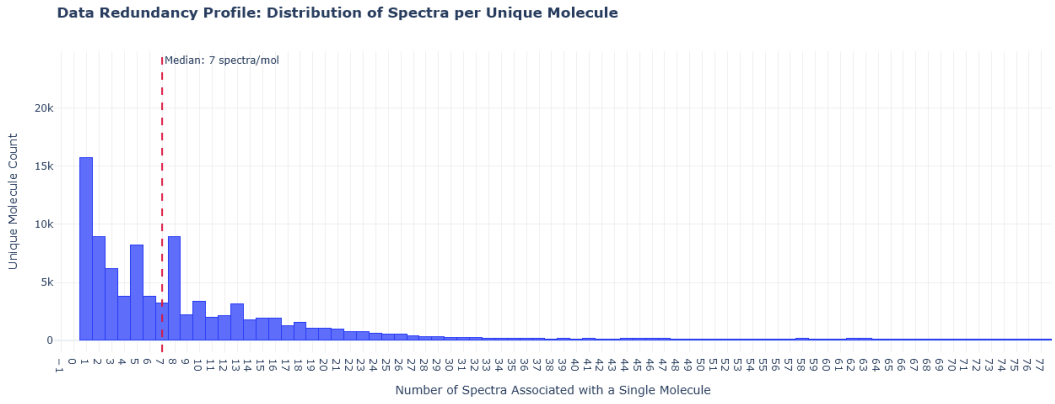


Figure 4. Distribution of spectra per unique molecule. The dashed line marks the median.

2. Sample Inspection

Figure 5 inspects a single molecule (precursor m/z 180.1019, [M+H]+, 199 recorded fragments). The 3D structure is generated from its SMILES string and placed beside the MS2 spectrum, with the base peak (m/z 110.0600) highlighted. This is mainly for end users, providing a 3D visualization of a molecule from its SMILES string to help them understand which parts of the molecule correspond to the mass spectrum.

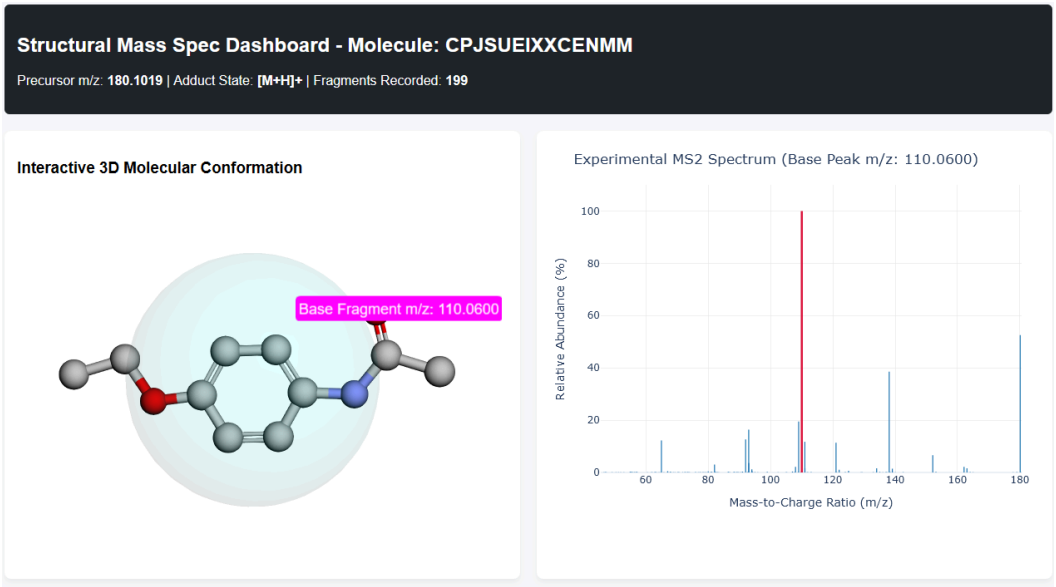


Figure 5. 3D molecule visualization (from SMILES) compared with its experimental MS2 spectrum.

3. Fragment-level Chemistry

Kendrick mass defect (Figure 6). Below roughly 300 m/z, fragments form a dense, upward-sloping band. This is the expected signature of small organic fragments rich in H, C, N, and O: each added hydrogen contributes about +0.0078 Da of mass defect relative to carbon. Above roughly 800 m/z the bins become sparse and noisy, so the model will see far fewer examples of large fragments.

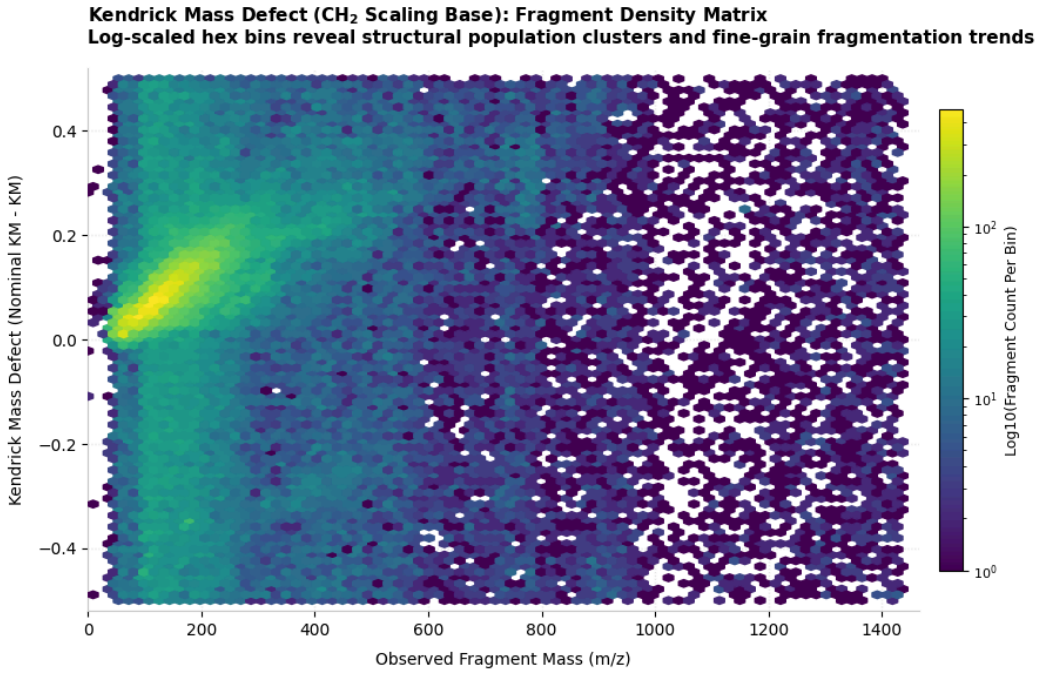


Figure 6. Kendrick mass defect (CH_2 scaling base) vs. observed fragment mass. Log-scaled hex bins show population clusters and fine-grained fragmentation trends.

Elemental mass defect (Figure 7). Most precursor masses show an increasing trend as mass increases, likely due to the addition of hydrogen atoms. A small group of spectra falls near a fractional mass of 1.0 around 300-400 m/z. These may contain heavier atoms such as chlorine or bromine. De novo sequence models often struggle with rare heavy atoms, so if these molecules matter for the task, adding explicit atom-level tokens or loss weighting are worth considering.

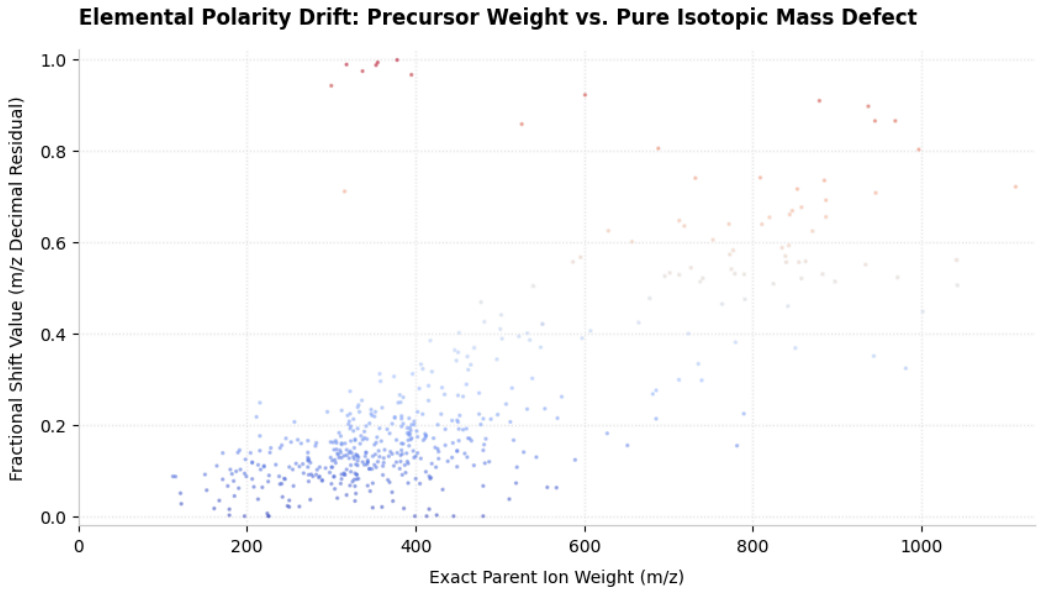


Figure 7. Precursor m/z vs. fractional mass defect (decimal residual).

4. Smiles Target Structure

Length vs. mass (Figure 8). The **normalized_smiles** text is translated token-by-token (e.g., using a Byte-Pair Encoding (BPE) tokenizer), so checking how text token length scales with parent molecule mass (precursor_mz) helps determine the transformer's MAX_LENGTH. This prevents sequences from clipping during training while keeping attention memory usage small.

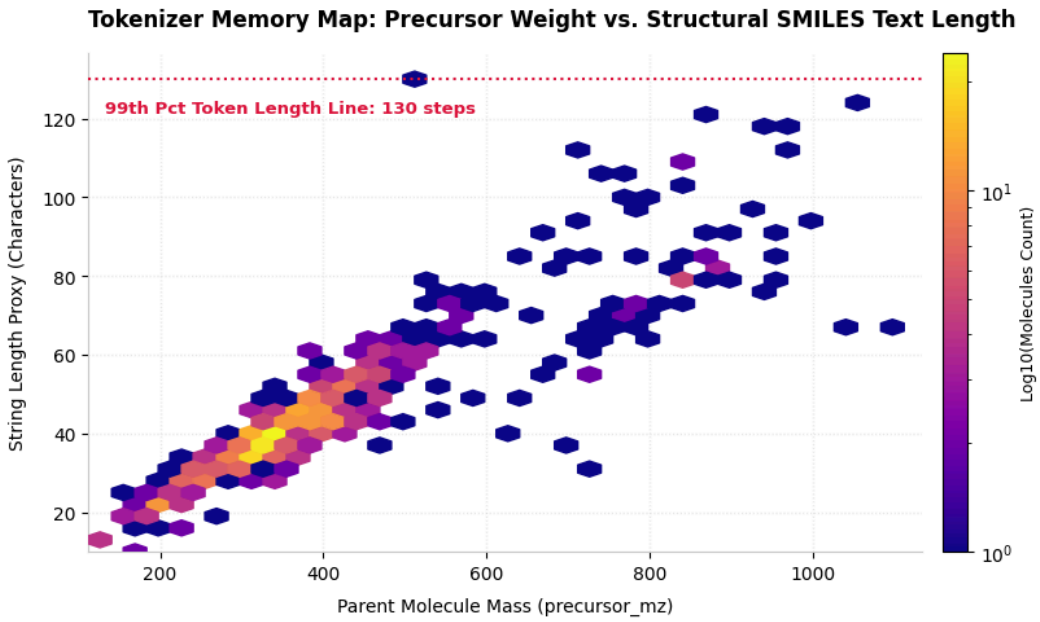


Figure 8. Precursor mass vs. SMILES string length (characters).

Structural Complexity (Figure 9). This visualization shows the frequency of structural characters, such as branching parentheses and ring closure number (1, 2, 3), across different precursor weights. These features typically increase with precursor mass, with most structures between 200-450 m/z having around 5-25 tokens. If high-mass regions don't show a linear increase in structural complexity tokens, the generator might suffer from "hallucination loop collapses" in heavy molecular zones.

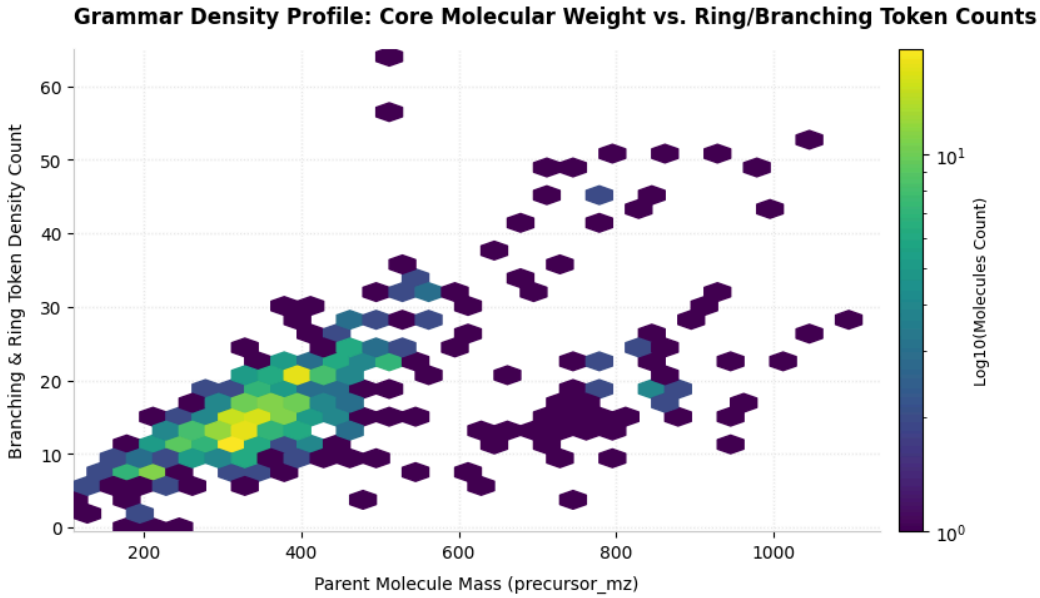


Figure 9. Precursor mass vs. ring and branching token counts.

5. Discussion of Missingness, Noise, and Quality Issues

Table 2. Data-quality issues and their implications

Issue	Evidence	Implication
Adduct imbalance	[M+H]+ = 53.1%, about 7 adducts at or below 0.2% (Figure 3)	Stratify folds, report per-adduct metrics, rare adducts are not reliably evaluable

6. Implications on Model Parameters

Spectrum EDA scanned 2,539,608 spectra and sampled 20,000 per split. At 128 spectral positions (127 fragment peaks plus precursor), about 90% of sampled spectra retain at least 98% of summed intensity (Figure 10). The implemented Transformer adds one metadata token: 129 total encoder positions. Its decoder limit is 160 tokens including BOS/EOS.

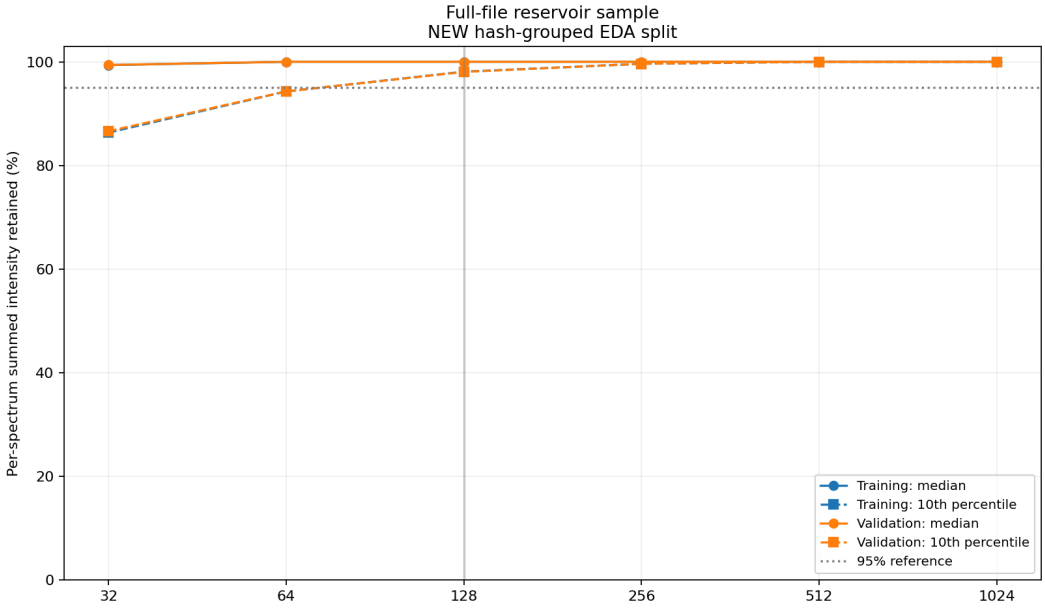


Figure 10. Per-spectrum intensity retained versus the spectral-position budget for training and validation samples. The plotted budget includes the precursor and excludes the separate metadata token. Thus 128 spectral positions correspond to 129 total Transformer encoder positions.

SMILES EDA checked decoder length and tokenizer coverage (Figure 11). Across **275,810 supplied molecular identities (220,770 training and 55,040 validation)**, token lengths had a median of **11**, a 99th percentile of **30**, and a maximum of **107**, with no unknown tokens among the selected representatives. This helps avoid oversized decoder sequences and accidental target truncation.

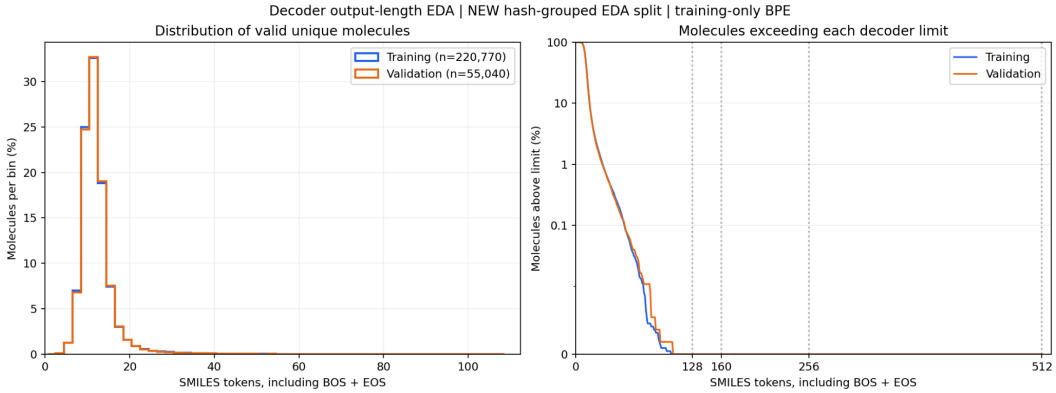


Figure 11. Decoder output-length EDA: token-length distribution (left) and percentage of molecules exceeding each limit (right), with training-only BPE.

Preprocessing plan

Preprocessing reads molecular columns in batches and caches normalization results; the current training loader still loads the full Parquet file.

Molecular Standardization and Identity-Based Data Partitioning

Molecules were validated with RDKit and normalized with competition-aligned tautomer handling, producing **274,193 normalized molecular identities** with one consistent SMILES target per identity.

All spectra belonging to the same identity were assigned to a single partition: generator training, validation, or ranker holdout. The saved audit found **zero identities shared across partitions**, preventing identical-molecule leakage. This reduces conflicting training targets and improves evaluation reliability while preserving all spectral measurements.

Table 3. Spectral records by partition.

Purpose	Spectral records
Train the Transformer generator	2,242,315
Validate the generator	49,951
Reserve for ranker training and evaluation	247,342
Total	**2,539,608**

Figure 11 reports the original supplied targets and historical 80/20 split. The benchmark uses competition-normalized targets, the saved three-way split, and a 160-token decoder cap.

Spectrum Cleaning and Filtering

The computational pilot removes nonfinite and nonpositive fragment m/z or intensity values, retains the 127 most intense fragment peaks, and includes one precursor position. Relative-intensity filtering below 1% of the base peak and removal of peaks above precursor mass remain proposed experiments; they were not used to produce the benchmark timings. Spectra with very few peaks will be investigated separately.

The retained spectrum therefore has at most 128 spectral positions. The Transformer also receives a separate metadata token, for a maximum of 129 total encoder positions. The EDA result that about 90% of sampled spectra preserve at least 98% of summed intensity applies to the 128-position spectral budget; it does not count metadata.

Segmentation and Input Representation

Each retained peak contains its m/z value and normalized intensity. Variable-length spectra are padded within each batch, and padding masks prevent padded positions from contributing as real spectral measurements. The Transformer embeds each mass/intensity pair and uses self-attention to relate fragment peaks, including peaks far apart on the m/z axis.

For the CNN-to-RNN baseline, the same retained peaks are placed on a fixed m/z grid before convolution. The pilot uses 0.1-m/z bins over [0, 4096), with separate channels for fragment intensities and a precursor indicator. This avoids applying local convolution to an intensity-sorted peak list. Masses outside the configured range produce an explicit error rather than silently being clipped.

The encoder representations differ by model: the Transformer has at most 129 input positions, whereas the CNN operates on two channels of 40,960 mass bins and then pools its features. Both generators share a training-only BPE tokenizer and a maximum of 160 SMILES tokens, including BOS and EOS. Longer targets are excluded explicitly; generated prefixes without EOS are rejected.

Normalization

Normalization will be applied to make the mass spectra more consistent across different measurements and improve model training stability. First, fragment intensities will be normalized relative to the highest-intensity peak in each spectrum, known as the base peak. This scales intensities to a common range between 0 and 1, preventing spectra with larger absolute intensity values from dominating training. We may also experiment with square-root or logarithmic intensity transformations to reduce the influence of extremely strong peaks while preserving information from weaker fragments. The m/z values will be represented using a consistent numerical scaling method. We will evaluate whether scaling m/z values by a fixed reference value or using training-set statistics produces better model performance. Any statistics learned during normalization will be calculated using only the training partition to prevent the data leakage. SMILES targets will also be standardized using the RDKit-based molecular normalization already described in our dataset partitioning process. The standardized SMILES strings will then be tokenized using the training-only BPE tokenizer. These normalization steps should improve numerical stability while preserving important chemical information.

Adduct and Collision-Energy Conditioning

Mass spectra can vary with the ion adduct and collision energy even when the underlying molecule is identical. Adducts such as [M+H]+ and [M+Na]+ shift the measured precursor m/z and may change fragmentation. The EDA found that [M+H]+ accounts for approximately 53.1% of spectra. Treating these conditions as inputs can help the model distinguish measurement effects from molecular-structure differences; a predictive benefit has not yet been established.

The computational pilot supplies adduct, ionization-mode and instrument categories and a collision-energy summary, with missingness handled explicitly. The Transformer combines these features in a separate metadata token, bringing its spectral-plus-metadata input cap to 129 positions. The CNN-to-GRU baseline instead adds metadata to its pooled context before initializing the recurrent decoder. Both generators use the same conditioning fields.

Collision energy governs the extent of fragmentation: higher energy often produces more and smaller fragments, while lower energy may preserve larger fragments. Available collision-energy measurements are summarized for conditioning; unavailable values remain explicitly missing. Stable categorical hashing avoids fitting a vocabulary on held-out data, but hash collisions are possible. Performance and per-adduct robustness must be measured on the identity-disjoint validation partition.

Augmentation

No data augmentation is currently planned. The dataset contains approximately 2.54 million spectra, and the first comparison will use existing measurements and the shared preprocessing pipeline. If validation reveals poor generalization for molecules or adducts with few examples, augmentation may be investigated later, subject to the remaining budget.

Model Plan

Build the planned CNN-to-RNN baseline and one Transformer encoder-decoder as the main model. The concrete pilot instantiates the RNN as a GRU. Pooled peak embeddings plus GRU is retained only as an auxiliary compute probe and is excluded from the required model comparison and budget. A ranker or ensemble is conditional on validated improvement and remaining resources.

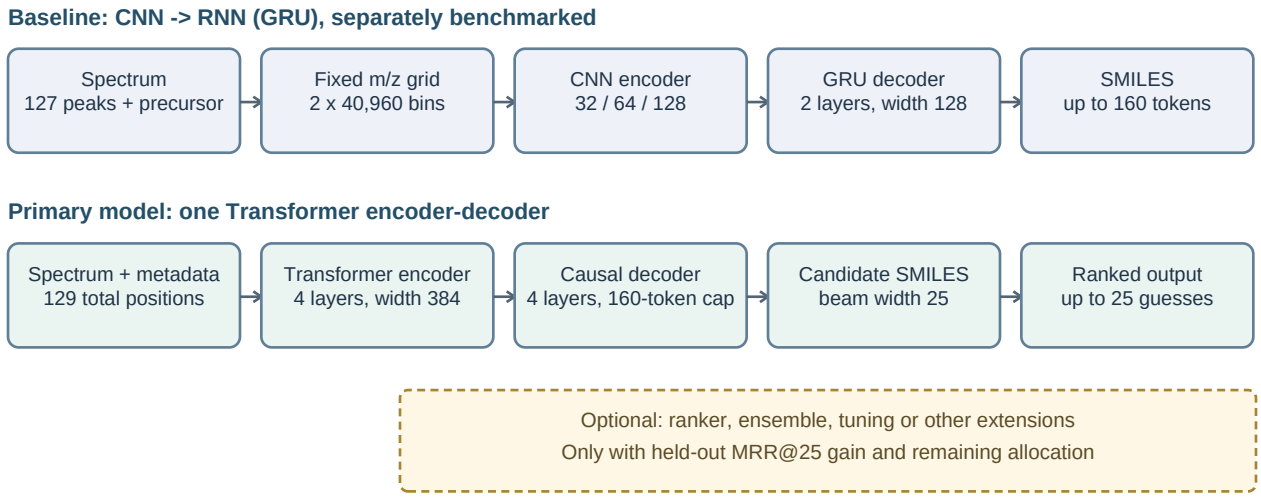


Figure 12. Planned CNN-to-RNN baseline and primary Transformer pipeline. The dashed box denotes optional extensions; it does not commit to an ensemble. Both generators use the same 160-token SMILES cap.

Stage	Model	What it does	Rationale
1	CNN -> RNN (GRU)	Convolves a fixed m/z grid; a recurrent decoder writes SMILES.	Required baseline. Separately measured; prediction quality remains untested.
2	Transformer encoder-decoder	Self-attention over 129 total encoder positions; causal decoding up to 160 tokens.	Relates distant fragments and conditions each output token on the spectrum.
3	Ranker / ensemble (optional)	Re-scores generated candidates or combines validated models.	Proceed only with held-out MRR@25 improvement and remaining GPU allocation.
4	Potential extensions	Tuning, attention variants or chemistry-informed training.	Defer until the two core models and evaluation work within budget.

1. Baseline: CNN -> RNN

The planned baseline remains a CNN-to-RNN sequence-to-sequence model. A CNN encodes local spectral patterns on a fixed m/z grid, and a GRU decoder generates SMILES one token at a time. This supplies a reference for evaluating whether the Transformer justifies its additional compute cost. Local convolution may miss fragment relationships spanning distant masses, while a recurrent hidden state may make long-range SMILES dependencies harder to represent; the comparison must be tested rather than assumed.

The measured pilot uses Conv1D channels 32, 64 and 128, kernel size 9 and stride 4 in each layer; adaptive average pooling to 16 cells; a 128-dimensional projection; and a two-layer width-128 GRU decoder. Metadata conditions the initial state. These are concrete baseline settings, not details specified by the original broad model plan. The separate CNN pilot measured 2,552 spectra/second; pooled-GRU timings are not substituted for it.

2. Proposed Model: Transformer Encoder-Decoder

The encoder receives 127 fragment peaks, one precursor position and one metadata token: 129 total encoder positions. The decoder generates BPE-tokenized SMILES with a limit of 160 tokens including BOS/EOS, using cross-attention to the encoded spectrum at each step. The pilot uses four encoder and four decoder layers, width 384, eight heads, feed-forward width 1536, dropout 0.1, and pre-LayerNorm.

Self-attention allows peaks to interact across m/z distances and can help relate fragments that originate from the same molecular substructure. Causal decoder attention can represent ring and branch dependencies while cross-attention conditions each token on the spectrum. The encoder processes peak representations in parallel. These mechanisms motivate the model; prediction quality and superiority over the baseline remain untested.

3. Optional Ranker or Ensemble

A single Transformer is the primary commitment. A ranker or additional generators are optional and require useful candidate recall, improvement in held-out MRR@25, and sufficient allocation after the evaluation/retry reserve. The initial ranker cost estimate covers at most 2,000 reserved structures with up to five spectra each (10,000 spectra), not all 247,342 reserved spectra. Generating candidates for that full partition would require about 10.01 GPU compute-hours, plus CPU processing and unmeasured full-holdout fitting.

4. Potential Extensions

Hyperparameter tuning, efficient attention, and chemistry-informed losses remain stretch goals. If an estimated additional cost exceeds the remaining budget, retain the single Transformer and defer the optional stage. Appendix A gives preliminary cost projections for the separately measured CNN baseline and Transformer.

Evaluation plan

The evaluation plan is to train the model and generate predictions on the Casimi Denovo Kaggle competition's test.parquet to find how well it scores. The model scoring and ranking is carried out by testing it on increasingly difficult tasks: known spectra, unknown spectra from known molecules, and new spectra whose molecule is entirely unknown. Models are scored based on whether the correct molecular structure appears among their top 25 predictions, with higher scores awarded when it ranks closer to the top.

Metrics

Primary metric: Mean Reciprocal Rank at 25 (MRR@25)

For each target molecule the model submits up to 25 candidate structures, ranked from best guess to worst. Rather than marking a single prediction as right or wrong, the score rewards how close to the top the first correct structure appears:

$$MRR@25 = \frac{1}{U}\sum_{u=1}^{U}\frac{1}{rank_u}$$

Where:

- U is the total number of unique molecules being evaluated in the test set.
- $rank_u$ is the position (from 1 to 25) of the **first correct chemical structure** predicted by the model for that molecule

For example, a correct prediction ranked first gets a score of 1.0, while one ranked second gets 0.5.

Since the same molecule can be represented by different SMILES strings due to tautomer variations or other equivalent representations, the competition evaluates molecular connectivity rather than requiring an exact string match. The grading pipeline uses RDKit to standardize predicted structures and convert them into InChIKeys. It then compares the first 14 characters of the InChIKeys (the connectivity block) to determine whether the predicted structure matches the target. This means differences in stereochemistry or tautomer form do not affect the score.

The overall MRR is calculated across three evaluation classes with different levels of difficulty:

Class	Representation in Test Set	Task	Typical Score
1: Library Reference Hits	~16%	Matching the test spectrum directly to an existing database spectrum.	Very high (often 0.90+ MRR).
2: Database candidates	~45%	The structure is in databases such as PubChem but has no spectrum; requires filtering exact-mass constitutional isomers.	Moderate
3: Novel natural products	~39%	Truly unseen structures; requires a de novo generator that writes SMILES from scratch	Very low (the main differentiator between top models)

Validation Strategy

Since our training data has around 2.5 million MS/MS spectra covering around 275k unique structures. It is considered large enough to use a fixed molecule-grouped split. We also have to keep the validation and test set untouched while training.

Split Design

1. Group rows by molecule identity
2. Assign whole group to train/validate/test
3. Handle the imbalance after grouping
4. Evaluate at the molecule level instead of spectrum

Imbalance Handling

For adduct imbalance, we plan to start with ordinary training on the natural distribution. First split train/validation data so each fold represents the adduct distribution as well as possible, and report a separate metric for each adduct alongside an overall metric. This shows whether the model works for common adducts and where it struggles on rare ones. If some adducts have only a handful of spectra, per-adduct validation metrics will be noisy. Report the sample count with each metric, and consider grouping very rare adducts into an “other” group for analysis. Keep the original adduct labels in the data if the model needs to predict or condition on them.

Leakage Awareness

Leakage would inflate validation scores without improving actual test performance so to prevent against this:

- The test set will not be used for any model fitting or selection.
- Techniques like hyperparameter tuning and early stopping will only be used on the validation set only, not the ranker or test data.
- Anything learned from data like the BPE tokenizer vocabulary will be fitted using only the training set.

Feasibility Risks and Mitigation

The short computational pilots support preliminary compute-feasibility estimates. They do not demonstrate prediction quality, full-dataset loading speed, or full-dataset host-memory feasibility. Both the CNN baseline and Transformer need structure-level validation before any claim about novel-molecule generalization.

Full-Dataset Loading and RAM

The timed loop cycles 4,096 prepared spectra. The original 2.19 GiB host-RAM peak belongs only to that small-sample PBS job, not a full pandas load of 2,539,608 spectra. The current training loader reads the full Parquet file into memory. Before a production run, profile the full load on an allocated compute node with a RAM limit, or implement streaming. The added 10% loading allowance is an assumption and cannot establish full-data behavior.

Representation, Rare Motifs and Generalization

Shared spectral motifs may help learning, but rare adducts, heavy atoms and large molecules can be poorly represented. Preserve identity-disjoint training, validation and ranker partitions; fit tokenization only on training targets; evaluate molecule-level MRR@25 and candidate recall; and report per-adduct sample counts so noisy subgroup metrics are identifiable. Use SMILES sequences as targets; InChIKey identity is used for matching and deduplication, not as a sequence target to interpolate.

Invalid or Missing Candidate Coverage

Autoregressive decoding may produce invalid or unfinished SMILES. Require EOS, parse with RDKit, normalize consistently and deduplicate by scoring identity. Measure valid unique coverage and candidate recall before adding a ranker, since ranking cannot recover a correct molecule absent from the generated candidates.

Budget Overrun and Optional Stages

The recalculated CNN + GRU/Transformer base plan is approximately 9.93 GPU allocation-hours for ten epochs per model plus the stated validation/test decoding. A proposed total of 12 GPU-hours reserves two additional hours for extra evaluation and partial retries. Including 0.060 hours already used, the minimum is 11.99 hours and only 0.01 hours remain outside the reserve. The capped ranker would require about 0.56 allocation-hours with inline CPU processing, so it is deferred under this proposal.

The confirmed 1.5-hour allowance applies to benchmarking only. A full-training allocation has not been specified or approved. The two-hour reserve cannot cover a full repeated seven-hour Transformer run; full retries or broader evaluation require a revised allocation. Appendix A records configurations, workload scopes, estimates and the underlying evidence bundle.

Appendix A. Computational Benchmark and Resource Plan

Evidence Status and Hardware

Prediction quality remains untested. The pilots support preliminary compute-feasibility estimates, not a full-dataset feasibility demonstration. One NVIDIA GeForce RTX 3090 was allocated per job using PyTorch 2.7.1+cu118 and CUDA 11.8, BF16 autocast with FP32 model/optimizer state, eight reserved CPU cores and four PyTorch/BLAS threads. CPU ranker fitting used one thread. The original successful job was 159855; the separate CNN job was 159856.

Configurations and Timing Procedure

Transformer: four encoder/four decoder layers, width 384, eight heads, feed-forward width 1536, dropout 0.1, pre-LayerNorm, vocabulary 512 and 17,614,208 parameters. Input: 129 total positions (127 fragments + precursor + metadata). Output: at most 160 SMILES tokens including BOS/EOS. CNN + GRU: two channels of 40,960 mass bins on [0, 4096) at 0.1 m/z, Conv1D channels 32/64/128, kernel 9, stride 4, pooling to 16 cells, a width-128 context and two GRU layers; 771,744 parameters. Metadata is fused into the recurrent initial state.

Each model used microbatch 32 and four-way accumulation (effective batch 128), 20 warm-up optimizer updates and 150 timed full updates, processing 19,200 spectra by cycling a 4,096-spectrum sample. Timing includes loading/tokenization, device transfer, forward pass, backward pass, gradient clipping and AdamW update; CUDA was synchronized around the interval. Dynamic padding was used. Sampled training target lengths had median 12, P95 24 and maximum 69 tokens. The separate full-cap check measures memory for one optimizer update, not sustained worst-case throughput.

Beam timing used 64 validation spectra, inference batch eight, beam width 25 and no KV cache. Candidate sampling used 16 reserved spectra, 100 samples per spectrum, inference batch eight and chunk 100. The same train-only tokenizer, normalized targets, identity partitions, peak retention and metadata were used. Recheck decoding after obtaining a validated checkpoint because generated length and candidate validity will change.

Loading and Memory Limitation

The full partition counts are 2,242,315 training, 49,951 validation and 247,342 reserved spectra. Timing a repeatedly reused 4,096-spectrum subset and observing 2.19 GiB peak host RAM in the original job does not establish full-Parquet loading behavior. Full-data RAM and I/O remain unmeasured. The current training loader reads the whole Parquet file; production use requires a compute-node full-load profile under a memory limit or streaming. GPU memory below is peak allocated memory; reserved-memory values remain available in the raw JSON records.

Memory is stage-specific: the CNN candidate-sampling stage recorded 0.37 GiB peak allocated and 10.77 GiB peak reserved GPU memory. Its 0.155 GiB training stress-check allocation does not bound inference reservation. Both figures come from the separate CNN raw record.

Appendix A. Resource Table

All total times are extrapolations; CPU fitting is a synthetic timing proxy. Alternative scopes replace their subset row and are not additive. Pooled-GRU is auxiliary and excluded from the base plan.

Stage	Priority	Hardware	Measured speed/memory	Estimated total time	Proceed/fallback condition
CNN + GRU (planned baseline) training	Required	RTX 3090	2551.8 spectra/s; 0.13 GiB allocated; 0.15 GiB at full caps (passed)	2.76 h estimated; 2,242,315 spectra x 10 epochs	Require a full-training allocation and held-out MRR@25 evaluation
Transformer encoder-decoder training	Required	RTX 3090	973.0 spectra/s; 0.86 GiB allocated; 1.16 GiB at full caps (passed)	7.16 h estimated; 2,242,315 spectra x 10 epochs	Require a full-training allocation and held-out MRR@25 evaluation
Pooled peak embeddings + GRU (auxiliary) training	Auxiliary only	RTX 3090	2877.2 spectra/s; 0.05 GiB allocated; 0.08 GiB at full caps (passed)	2.43 h estimated; 2,242,315 spectra x 10 epochs	Auxiliary measurement only; excluded from the planned baseline and budget
CNN + GRU beam validation	Base plan	RTX 3090	290.25 spectra/s; 0.06 GiB allocated; beam 25	0.00053 GPU h (~1.9 s); 553 validation spectra	Evaluate the baseline and Transformer on the same held-out identities
Beam decoding: validation subset + test	Base plan	RTX 3090	47.82 spectra/s; 0.28 GiB allocated; beam 25	0.0103 GPU h (~37 s); 553 validation + 1,213 test spectra	Remeasure on validated weights; lower beam width or inference batch if needed
Beam decoding: full validation + test	Alternative scope	RTX 3090	Same 47.82 spectra/s extrapolated from 64 measured spectra	0.297 GPU h (~17.8 min); 49,951 validation + 1,213 test spectra	Replaces the subset beam row; additional evaluation passes need additional budget
Ranker generation: capped subset (<=10,000 spectra)	Optional	RTX 3090	6.86 spectra/s; 100 samples/spectrum	0.40 GPU compute h + 0.14 CPU chemistry h; 2,000 structures x up to 5 spectra	Only if candidate recall is useful and held-out ranker MRR@25 improves; remeasure with trained weights
Ranker generation: all 247,342 reserved spectra	Deferred alternative	RTX 3090	Same 6.86 spectra/s extrapolated from 16 measured spectra	10.01 GPU compute h + 3.39 CPU chemistry h; full-holdout fitting unmeasured	Replaces the capped-subset row; requires a separate allocation and trained-checkpoint timing
Ranker fitting + refit	Optional	CPU, one thread	58.8 s synthetic proxy; 1,000,000 candidate rows	0.016 CPU h proxy; 2,000 groups x 500 candidates/group	Proxy for the capped subset; cannot transfer to full-holdout fitting; require validation gain
Ensemble	Deferred	RTX 3090	Unmeasured	Unestimated; no extra model committed	If estimated additional cost exceeds remaining budget, retain the single Transformer; otherwise require held-out MRR@25 gain

The base plan uses the separately measured CNN + GRU baseline and one Transformer. GPU compute time excludes CPU processing; GPU allocation time includes CPU work when the GPU remains reserved. Prediction quality remains untested.

Appendix A. Cost Assumptions and Decision Rule

Training Projection

Training hours = training spectra x planned epochs / measured spectra per second / 3600. The scenario uses 2,242,315 spectra x ten epochs per required model, plus measured validation and checkpoint costs, a 10% additional loading allowance and preparation time. It predicts 2.76 hours for CNN + GRU and 7.16 hours for the Transformer. These are extrapolations from short pilots; no full training has been completed.

Validation Scope

The training estimates include 97 teacher-forced token-loss validation events, capped at 5,000 spectra each and extrapolated from the 553-spectrum sample. The base plan separately includes one beam evaluation of the same 553 spectra for each required model and Transformer decoding of the 1,213 test spectra. It does not include beam decoding the full 49,951-spectrum validation partition after every epoch. One full Transformer validation-plus-test beam pass would take about 0.297 GPU-hours at the measured rate.

Ranker Scope and Allocation Accounting

The 0.40 GPU compute-hour ranker estimate covers at most 2,000 structures x five spectra = 10,000 spectra, plus about 0.14 CPU chemistry-hours. Sampling all 247,342 reserved spectra would instead require about 10.01 GPU compute-hours and 3.39 CPU chemistry-hours; full-holdout fitting is unmeasured. The 58.8-second CPU fixture fits 70% and then all of 2,000 synthetic groups with 500 candidates each, 250 trees and 15 leaves; its synthetic labels provide no quality evidence. Retaining the GPU through subset CPU chemistry and this fitting proxy would cost about 0.56 allocation-hours. Full-holdout inline generation would cost about 13.42 allocation-hours before fitting.

Actual Allowance and Proposed Training Budget

Only the measurement allowance of 1.5 hours is confirmed. GPU allocation usage across the original job, its startup retry and the CNN job totals 216 seconds (0.060 hours). The corrected CNN + GRU/Transformer base plan is 9.93 hours. Adding a two-hour reserve for extra evaluation/partial retries and the recorded measurement use gives 11.99 hours. A proposed 12-hour total leaves 0.01 hours outside the reserve; this is a proposal, not an approved full-training allocation.

If an optional stage costs more than the remaining allocation, retain the single Transformer and defer that stage. The capped ranker does not fit the proposed spare budget; an ensemble is unmeasured and deferred. Any later ranker or ensemble also requires improved held-out MRR@25. The earlier 9.60/11.65-hour totals included pooled-GRU, not the planned CNN baseline, and are superseded.

Supporting Evidence

The accompanying benchmark_evidence.zip supplies raw model JSON records, hardware and sample manifests, PBS output/accounting, package versions, source snapshots and checksums. A labeled copy clarifies the CNN raw record's inherited peak-position counter: the CNN actually uses 40,960 grid bins per channel. Timing values are unchanged. The original input PDF and EDA plots remain preserved; figures on pages 8 and 9 are historical EDA, not prediction-quality results.